Laplace

L (F(t))	$L^{-1}(F(t))$
C	$\frac{C}{S}$
e^{at}	$\frac{1}{S-a}$
t^n	$\frac{n!}{S^{n+1}}$
$\sin(at)$	$\frac{a}{S^2+a^2}$
$\cos(at)$	$\frac{S}{S^2+a^2}$
$\sinh(at)$	$\frac{a}{S^2-a^2}$
$\cosh(at)$	$\frac{S}{S^2-a^2}$

- $L\big(af(t)\big) = a\,L\,\big(f(t)\big)$
- $L\big(f(t) \pm g(t)\big) = L\big(f(t)\big) \pm L\big(g(t)\big)$

Ex:- $f(t) = sin^2(3t) + 5$

$$f(t) = \left(\frac{1}{2} - \frac{1}{2}\cos(6t)\right) + 5 = L\left(\frac{1}{2}\right) - \frac{1}{2}L(\cos(6t)) + L(5)$$

$$f(s) = \frac{1}{2S} - \frac{1}{2}\left(\frac{S}{S^2 + 36}\right) + \frac{5}{S}$$

Ex:- $f(s) = \frac{4}{S+3} - \frac{4}{S^2-12}$

$$L^{-1}\left(\frac{4}{S+3} - \frac{4}{S^2-12}\right) = 4\,L^{-1}\left(\frac{1}{S+3}\right) - \frac{4}{\sqrt{12}}L^{-1}\left(\frac{\sqrt{12}}{S^2-12}\right)$$

$$f(t) = 4e^{-3t} - \frac{4}{\sqrt{12}}\sinh(\sqrt{12}\,t)$$

Ex:- $f(s) = \frac{2S+1}{S^2-5} - \frac{7S+3}{S^2+1}$

$$= \frac{2S}{S^2 - 5} + \frac{1}{S^2 - 5} + \frac{7S}{S^2 + 1} + \frac{3}{S^2 + 1}$$

$$= 2L^{-1}\left(\frac{S}{S^2 - 5}\right) + L^{-1}\left(\frac{1}{S^2 - 5}\right) + 7L^{-1}\left(\frac{S}{S^2 + 1}\right) + 3L^{-1}\left(\frac{1}{S^2 + 1}\right)$$

$$f(t) = 2\cosh(\sqrt{5}t) + \frac{1}{\sqrt{5}}\sinh(\sqrt{5}\,t) + 7\cos(t) + 3\sin(t)$$

1) First Shift:

- $e^{at} f(t) \Rightarrow s \to s - a \mid s - a \to s \Rightarrow e^{at} f(t)$

Ex:- $f(t) = e^{-3t} \cos^2(6t)$

$$= \cancel{e^{-3t}} \left(\frac{1}{2} + \frac{1}{2}\cos(12t)\right) = L\left(\frac{1}{2}\right) + \frac{1}{2} L(\cos(12t))$$

$$f(s) = \frac{1}{2S} + \frac{1}{2}\left(\frac{S}{S^2+144}\right) \quad S \to S - (-3) = S + 3$$

$$f(s) = \frac{1}{2(S+3)} + \frac{1}{2}\left(\frac{S+3}{(S+3)^2 + 144}\right)$$

Ex:- $f(s) = \frac{S}{S^2+10S+8}$

$$= \frac{S}{\left(S + \frac{10}{2}\right) + 25 - 17} = \frac{S+5-5}{(S+5)^2 - 17}$$

$$= \frac{S+5}{(S+5)^2 - 17} - \frac{5}{(S+5)^2 - 17} \qquad S + 5 \to S \Rightarrow e^{-5t}$$

$$= \frac{S}{S^2 - 17} - \frac{5}{S^2 - 17} = L^{-1}\left(\frac{S}{S^2 - 17}\right) - \frac{5}{\sqrt{17}} L^{-1}\left(\frac{\sqrt{17}}{S^2 - 17}\right)$$

$$f(t) = e^{-5t}\left(\cosh\left(\sqrt{17}t\right)\right) - \frac{5}{\sqrt{17}} \sinh\left(\sqrt{17}\, t\right)$$

2) Diffrtential Equation Using Laplace Transform:-

$$L\big(F(t)\big) = F(s)$$

$$L\big(F'(t)\big) = S^1F(s) - F(0)$$

$$L\big(F''(t)\big) = S^2F(s) - S^1F(0) - F'(0)$$

Ex:-

$y'' + 4y' + 3y = 0 \qquad y(0) = 0\,, y'(0) = 1$

$$L(y'') + 4L(y') + 3L(y) = L(0)$$

$$(S^2y(s) - Sy(0) - y'(0)) + 4(Sy(s) - y(0)) + 3(y(s)) = 0$$

$$y(0) = 0\,, y'(0) = 1$$

$$(S^2y(s) - S \times 0 - 1) + 4(Sy(s) - 0) + 3(y(s)) = 0$$

$$(S^2 + 4S + 3)y(s) = 1$$

$$y(s) = \frac{1}{S^2 + 4S + 3} = \frac{1}{(S+3)(S+1)}$$

$$= \frac{A}{S+3} + \frac{B}{S+1} = L^{-1}\left(\frac{A}{S+3}\right) + L^{-1}\left(\frac{B}{S+1}\right)$$

$$y(t) = Ae^{-3t} + Be^{-t} = \frac{-1}{2}e^{-3t} + \frac{1}{2}e^{-t}$$

$$A = \lim_{s \to -3} \frac{1}{S+1} = \frac{-1}{2}$$

$$B = \lim_{s \to -1} \frac{1}{S+3} = \frac{1}{2}$$

2) Rules:-

1) $L\left(\int_0^t f(\tau)\ d\tau\right) = \frac{f(s)}{s}$ هات لابلاس للدالة جوه التكامل

Ex:- $F(t) = \int_0^t \sin(\tau)\, d\tau$

$$L\left(\int_0^t \sin(\tau)\, d\tau\right) = \frac{1}{s}\left(\frac{1}{S^2+1}\right)$$

$$F(s) = \frac{1}{S(S^2+1)}$$

Ex:- $F(t) = \int_0^t 5\,\tau^4 + 1\, d\tau = L\left(\int_0^t 5\,\tau^4 + 1\, d\tau\right)$

$$F(s) = \frac{1}{s}\left(5\left(\frac{4!}{S^5}\right) + \frac{1}{s}\right)$$

Ex:- $F(t) = \int_0^t e^{-2\tau}\cos(2\tau)\ d\tau = L\left(\int_0^t e^{-2\tau}\cos(2\tau)\ d\tau\right)$

$$S \to S+2 \ \ for \cos only$$

$$F(S) = \frac{1}{s}\left(\frac{S}{S^2+4}\right) = \frac{1}{s}\left(\frac{S+2}{(S+2)^2+4}\right)$$

Ex:- $F(t) = e^{-2\tau}\int_0^t \cos(2\tau)\ d\tau = L\left(e^{-2\tau}\int_0^t \cos(2\tau)\ d\tau\right)$

$$S \to S+2 \ \ for\ All$$

$$F(S) = \frac{1}{s}\left(\frac{S}{S^2+4}\right) = \frac{1}{S+2}\left(\frac{S+2}{(S+2)^2+4}\right)$$

$$L^{-1}\left(\frac{f(s)}{s}\right) = \left(\int_0^t f(\tau)d\tau\right)$$

هات لابلاس انفرس ل f(s)

شيل 1 علي S و حط التكامل و هات ناتج التكامل

Ex:- $F(S) = \frac{1}{S}\left(\frac{1}{S+3}\right) = L^{-1}\left(\frac{1}{S}\left(\frac{1}{S+3}\right)\right)$

$$= \int_0^t e^{-3\tau}d\tau = \frac{e^{-3\tau}}{-3} = -\frac{1}{3}(e^{-3t} - 1)$$

Another Solution:

$$L^{-1}\left(\frac{1}{S}\left(\frac{1}{S+3}\right)\right) = L^{-1}\left(\frac{1}{S(S+3)}\right)$$

$$L^{-1}\left(\frac{A}{S} + \frac{B}{S+3}\right) \rightarrow F(t) = A + Be^{-3t} = \frac{1}{3} - \frac{1}{3}e^{-3t}$$

$$A = \lim_{s\to 0}\frac{1}{S+3} = \frac{1}{3}$$

$$B = \lim_{s\to -3}\frac{1}{S} = \frac{-1}{3}$$

Ex:- $F(s) = \frac{1}{s}\left(\frac{S}{S^2-9}\right) = L^{-1}\left(\frac{1}{S}\left(\frac{S}{S^2-9}\right)\right) = L^{-1}\left(\frac{\cancel{S}}{\cancel{S}(S^2-9)}\right)$

$$\frac{1}{3} \times L^{-1}\left(\frac{1\times 3}{S^2-9}\right) \rightarrow F(t) = \frac{1}{3}\sinh(3t)$$

Ex:- $F(s) = \frac{S+2}{S^3+4S^2+5S} = L^{-1}\left(\frac{S+2}{S^3+4S^2+5S}\right)$

$$L^{-1}\left(\frac{S+2}{S(S^2+4S+5)}\right) = L^{-1}\left(\frac{S+2}{S((S+2)^2+1)}\right)$$

$$L^{-1}\left(\frac{\cancel{S+2}}{S((\cancel{S+2})^2+1)}\right) = L^{-1}\left(\frac{S}{S((S)^2+1)}\right) = L^{-1}\left(\frac{1}{S}\left(\frac{S}{S^2+1}\right)\right)$$

$$F(t) = \int_0^t e^{-2\tau}\cos(\tau)d\tau$$

---

2) $L\left(t^n\ F(t)\right) = (-1)^n \frac{d^n}{ds^n}\ F(s)$ شوف الأس كام واشتق بعددهم

Ex:- $F(t) = t\sin(t)\ n = 1$

$$L(\sin(t)) = (-1)^1\ \frac{d^1}{ds^1}\ \frac{1}{S^2+1} = \frac{d}{ds}\ (-(S^2+1)^{-1})$$

$$= -\left(\frac{-2S}{(S^2+1)^2}\right) \quad \rightarrow \quad F'(S) = \frac{2S}{(S^2+1)^2}$$

Ex:- $F(t) = t^7e^{3t}$ ~~$n = 7$~~

$$L(t^7e^{3t}) \rightarrow \quad S \rightarrow S-3 \quad \rightarrow \quad F(S) = \frac{7!}{(S-3)^8}$$

Ex:- $F(t) = t^2\cos(t) \quad n = 2$

$L(\cos(t))$

$$F(S) = \frac{d^2}{ds^2}\left(\frac{S}{S^2+1}\right) = \frac{(S^2+1)-2S^2}{(S^2+1)^2}$$

$$= \frac{(S^2+1)^2(2S-4S) - \left((S^2+1)-2S^2\right)(2(S^2+1)2S)}{((S^2+1)^2)^2}$$

$L^{-1}\big(F'(S)\big) = -t\,F(t)$ هات لابلاس انفرس للمشتقة

Ex:- $F(s) = \ln(\frac{S}{S^2+1})$

$$F(S) = \ln(S) - \ln(S^2+1)$$

$$F'(S) = \frac{1}{S} - \frac{2S}{S^2+1}$$

$$L^{-1}\big(F'(S)\big) = L^{-1}\left(\frac{1}{S} - \frac{2S}{S^2+1}\right)$$

$$-t\,F(t) = 1 - 2\cos(t) \qquad \div -t$$

$$F(t) = \frac{-1}{t} + \frac{2\cos(t)}{t}$$

Ex:- $F(s) = tan^{-1}\left(\frac{S}{2}\right)$

$$F'(S) = \frac{1}{1+\frac{S^2}{4}} \times \frac{1 \times 2}{2 \times 2} = \frac{2}{S^2+4}$$

$$L^{-1}\big(F'(S)\big) = L^{-1}\left(\frac{2}{S^2+4}\right)$$

$$-t\,F(t) = \sin(2t) \qquad \div -t$$

$$F(t) = -\frac{\sin(2t)}{t}$$

3) $L\left(\frac{F(t)}{t}\right) = \int_S^\infty F(z)dz$

Ex:- $F(t) = \frac{\sin(t)}{t}$

$$L\left(\frac{\sin(t)}{t}\right) = \int_S^\infty \frac{1}{z^2+1} dz = tan^{-1}(Z)\infty, S$$

$$= tan^{-1}(\infty) - tan^{-1}(S)$$

$F(s) = \frac{\pi}{2} - tan^{-1}(S)$ | $tan^{-1}(\infty) = \frac{\pi}{2}$

---

4) Second Shift :

$L\big(u_a(t)F(t-a)\big) = e^{-as}L(F(t))$ تأكد إنه نفس الرقم

نفس الإشارة a

Ex:- $F(t) = u_2(t)\sin(t-2)$

$L(u_2(t)\sin(t-2)) = e^{-2S}\,L(\sin(t))$

$F(s) = e^{-2S}\left(\frac{1}{S^2+1}\right)$

Ex:- $F(t) = u_5(t)\,(t)$

$L(u_5(t)\,(t-5)+5\,) = e^{-5S}L(t+5)$

$F(S) = e^{-5s}\left(\frac{1!}{S^2} + \frac{5}{S}\right)$

Ex:- $F(t) = u_3(t)e^t \cos(t-3) + u_6(t)$ First & Second Shift

$$L(u_3(t)e^t \cos(t-3) + u_6(t))$$
$$= L(u_3(t)e^t \cos(t-3)) + L(u_6(t))$$
$$= L(u_3(t)e^{t-3+3} \cos(t-3)) + L(u_6(t)\,(t-6)^0)$$
$$= e^3\, L(u_3(t)e^{t-3} \cos(t-3)) + L(u_6(t)\,(t-6)^0)$$

$$F(S) = e^3\left(e^{-3S}\left(\frac{S}{S^2+1}\right)\right) + e^{-6S}\left(\frac{1}{S}\right)$$

$S \to S-1 \quad \to\to \quad \uparrow$

$$F(S) = e^3\left(e^{-3S}\left(\frac{S-1}{(S-1)^2+1}\right)\right) + e^{-6S}\left(\frac{1}{S}\right)$$

First Shift : e^{-at} $S \to S+a$ a عكس إشارة

Second Shift: e^{-as} $t \to t-a$ a نفس إشارة

$$L^{-1}\left(e^{-as}\, f(s)\right) = u_a(t)F(t-a)$$

Ex:- $L^{-1}\left(e^{-4S}\left(\frac{S-1}{S^2+6S-9}\right)\right)$

$$L^{-1}\left(\frac{S-1}{(S+3)^2-18}\right) = L^{-1}\left(\frac{S+3-3-1}{(S+3)^2-18}\right)$$
$$= L^{-1}\left(\frac{S+3}{(S+3)^2-18}\right) - \frac{1}{\sqrt{18}}L^{-1}\left(\frac{4\times\sqrt{18}}{(S+3)^2-18}\right)$$

$$F(t) = e^{-3t}\left(\cosh\left(\sqrt{18}t\right) - \frac{4}{\sqrt{18}}\left(\sinh\left(\sqrt{18}t\right)\right)\right.$$

$$F(t) = u_4(t)e^{-3t-4}\left(\cosh\left(\sqrt{18}t-4\right) - \frac{4}{\sqrt{18}}\left(\sinh\left(\sqrt{18}t-4\right)\right)\right.$$